

LinkedIn AI Post Automation

A Beginner-Friendly, Step-by-Step Setup Guide

Turn your own notes into polished LinkedIn posts automatically. Gemini AI writes the draft, a built-in validator checks the quality, and the official LinkedIn API publishes it - all on a schedule you choose.

This guide assumes nothing: every command, every click, and every value is explained. Work through the steps in order - you should be posting automatically within an hour. Stuck? Jump straight to the **Troubleshooting** section at the end.

■ **100% Free:** This entire workflow costs **\$0** - no paid plans, no credit card, no hidden fees. GitHub Actions gives free automation minutes, Gemini AI has a generous free tier for post generation, and publishing through the LinkedIn API is free. The only cost is a few minutes of your time to set it up once.

What This Project Does (In Plain English)

- You keep a short notes file (`today_notes.txt`) - just a few sentences of what you learned or did.
- Gemini AI (Google's free assistant) turns those notes into a natural, ready-to-publish LinkedIn post.
- A **self-review validator** checks the draft against your rules - length, tone, hashtags, emojis - and rewrites it if needed.
- The post is published to your LinkedIn profile through the **official LinkedIn API** (your real account, real post).
- A **GitHub Actions workflow** runs all of this automatically on a schedule - no computer needs to stay on.

Where everything runs:

Piece	What it is	Where it runs
Node.js scripts	The brains: generate, validate, publish	Your computer (tests) & GitHub (schedule)
Gemini API	Free AI that writes the posts	Google's cloud
LinkedIn API	Publishes to your profile	LinkedIn's cloud
GitHub Actions	Runs the automation on a timer	GitHub's cloud

Before You Begin - What You Need

Everything below is free. Two small installs (Node.js and Git) and three online accounts. If you already have any of them, just skip ahead.

Tool	Where to get it	Why you need it
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Node.js 18+ (Node 20 recommended)	nodejs.org	Runs the automation scripts
Git	git-scm.com	Downloads the code; saves your changes
GitHub account	github.com	Hosts the scheduled workflow
Gemini API key (free)	aistudio.google.com	AI post generation
LinkedIn Developer App	linkedin.com/developers	Publishing access to your account

Quick check: open a terminal (Command Prompt / PowerShell / Terminal) and run `node --version` and `git --version`. If each prints a version number, you are ready.

The Complete Beginner Walkthrough - Zero to First Post

This section is for absolute beginners: if you have never opened a terminal, never used GitHub, and never touched a settings file - start here. Follow it **click by click** and **command by command**. You don't need to understand every detail, just follow the order. When you finish, your LinkedIn posts will publish themselves on a schedule.

Words you will meet along the way:

Word	What it really means
Terminal	A window where you type commands and press Enter instead of clicking buttons
Clone	Download a copy of the project's code to your computer
Fork	Your own personal copy of a project on GitHub
cd	Change directory - move into a folder
npm	A helper tool that installs the project's libraries
.env	Your private settings file - never shared or uploaded

PART 1

Install the two tools (10 minutes)

- **Node.js:** go to **nodejs.org** and click the big green **LTS** download button. Open the downloaded file and click Next, Next, Finish (Windows) or drag it into Applications (Mac).
- **Git:** go to **git-scm.com**, click the download button for your system, and install with the default settings.

Now check they work. Open your terminal and type each line, pressing Enter after it:

```
node --version git --version
```

You should see a version number under each command - something like `v20.12.0` for Node and `git version 2.45.0` for Git.

■ **Tip:** Where is the terminal? Windows: press the Start button and type "Command Prompt" - or open "Git Bash" (installed together with Git). Mac: press Cmd+Space, type "Terminal" and press Enter.

PART 2

Open GitHub and copy the code link (5 minutes)

- Open your browser, go to **github.com**, and sign in (create a free account if you don't have one).
- Open the project page: **github.com/your-username/linkedin-ai-post-automation** (replace your -username with the owner's username).
- Click the **Fork** button at the top right. Wait a few seconds while GitHub makes your personal copy - a fork is *your* project now.
- On your fork, click the green **<> Code** button, make sure the **HTTPS** tab is selected, then click the small copy icon (two overlapping squares) next to the address.

The link you just copied looks like this:

```
https://github.com/your-username/linkedin-ai-post-automation.git
```

PART 3

Download the code in the terminal (5 minutes)

Go back to your terminal and type these three lines one at a time, pressing Enter after each:

```
git clone https://github.com/your-username/linkedin-ai-post-automation.git cd
linkedin-ai-post-automation npm install
```

- `git clone ...` downloads the code - you'll see files flying by ("Receiving objects").
- `cd ...` moves you inside the project folder (check the folder name appears in the prompt).
- `npm install` downloads the helper libraries. It can take a minute - wait until it finishes.

■■■ **Important:** Wait until each command finishes and you get your prompt back before typing the next one. If npm shows errors, you are probably on an old Node version - reinstall Node.js LTS and retry.

PART 4

Create your keys and connect your account (15 minutes)

1. Gemini API key (the AI): go to **aistudio.google.com/app/apikey**, click **Create API key**, and copy it (keys start with AIza...).

2. LinkedIn app (publishing): go to **linkedin.com/developers/apps** -> **Create app**, request the **Share on LinkedIn** product, and add this redirect URL: **http://127.0.0.1:8585/callback**. Copy your **Client ID** and **Client Secret**.

3. Make your private settings file: in the terminal type:

```
cp .env.example .env
```

4. Open the file and fill it in. Windows: type `notepad .env`. Mac: type `open -e .env`. Replace the four example values with your real ones (Gemini key, Client ID, Client Secret) and save.

5. Connect your LinkedIn account: type `npm run auth`. A URL prints - open it in your browser, approve the **w_member_social** permission, then copy the **entire** address from the address bar and paste it back into the terminal, press Enter.

The script prints `LINKEDIN_ACCESS_TOKEN=...` - add that line to your `.env` file and save. Done - your computer is connected.

■■■ **Important:** Your LinkedIn token expires in about 60 days. When it does, just run `npm run auth` again. Never share the token or your Client Secret.

PART 5

Test everything on your computer (5 minutes)

Run these three commands in order:

```
npm test npm run dry-run npm run publish
```

- `npm test` - runs the built-in tests. All should pass (green).
- `npm run dry-run` - generates a sample post and prints it. **Nothing is published.**
- `npm run publish` - publishes one real post. Check your LinkedIn profile - it's there!

■ **Tip:** Dry-run is your safety net: use it whenever you change your notes or rules. It shows the exact post that would be published - without publishing it.

PART 6

Upload to GitHub and turn on the schedule (10 minutes)

1. Create your repository: on github.com click the + icon (top right) -> **New repository**. Name it `linkedin-ai-post-automation`, keep it **Private** for now, and click **Create repository**. Do NOT tick "Add a README" - your project already has files.

2. Connect and upload your code: in the terminal type (using *your* link):

```
git remote add origin https://github.com/your-username/linkedin-ai-post-automation.git  
git push -u origin main
```

3. Give the cloud its keys: on GitHub go to **Settings** -> **Secrets and variables** -> **Actions** -> **New repository secret**. Add `GEMINI_API_KEY` (your key) and `LINKEDIN_ACCESS_TOKEN` (your token) - exactly as in your `.env` file. Click **Add secret** after each.

4. Run it once: open the **Actions** tab -> **LinkedIn AI Post** -> **Run workflow** -> **Run workflow**. Watch the steps turn green.

- Green checkmarks = success. Your first scheduled post pipeline is live!
- The schedule in `.github/workflows/linkedin-post.yml` now posts automatically on your chosen days.

■ **Tip:** If the push asks you to log in, a GitHub browser window pops up - approve it. That's normal and only happens the first time.

That's it - you're live! From now on, before each posting day you only do one thing: write a few lines of notes into `config/today_notes.txt`. The computer handles the rest. The sections below are your detailed reference - for changing topics, hashtags, post length, the schedule, and going public.

Step 1 - Get a Free Gemini API Key

STEP 1

Create your Google AI key

- Go to aistudio.google.com/app/apikey and sign in with a Google account.
- Click **Create API key** (accept the terms first if prompted).
- Copy the key - Gemini keys always start with AIza....
- Store it somewhere safe; you will need it again in Steps 4 and 7.

```
# Example value (yours will be unique) GEMINI_API_KEY=your_gemini_api_key_here
```

■ **Tip:** The free tier is easily enough for a few posts per week - no credit card required.

Step 2 - Set Up a LinkedIn Developer App

STEP 2

Register your app

- Go to linkedin.com/developers/apps and click **Create app**.
- Give it a name and a page; agree to the terms.
- Under **Products**, request the **Share on LinkedIn** product - this is what allows posting.
- Under **Auth**, add exactly this redirect URL:

```
http://127.0.0.1:8585/callback
```

- Copy your **Client ID** and **Client Secret** - you will use them in Step 4.

```
# Example values (yours will be unique) LINKEDIN_CLIENT_ID=your_linkedin_client_id_here  
LINKEDIN_CLIENT_SECRET=your_linkedin_client_secret_here
```

■■ **Important:** Keep the Client Secret private - anyone with it can act as your app. If it is ever shared or leaked, click "Generate new secret" in your app settings immediately.

■ **Tip:** New apps are sometimes restricted from the "Share on LinkedIn" product for a short period. If it is not approved yet, every feature still works except publishing (use dry-run mode in Step 6 while you wait).

Step 3 - Get the Code

STEP 3

Copy the project to your computer

Open a terminal in the folder where you keep projects, then run:

```
git clone https://github.com/your-username/linkedin-ai-post-automation.git cd  
linkedin-ai-post-automation npm install
```

What these commands do:

- `git clone` downloads the project (replace your -username with the owner's GitHub username).
- `cd` moves you into the project folder.
- `npm install` downloads the project's helper libraries - run it exactly once.

■ **Tip:** Prefer to publish your own version? Click **Fork** on the GitHub page first, then clone <https://github.com/<your-username>/linkedin-ai-post-automation.git>.

Step 4 - Configure the Project

STEP 4

Create your .env file

Copy the template and open it in any text editor (VS Code, Notepad, etc.):

```
cp .env.example .env
```

Fill in these four values:

Variable	What to put here
GEMINI_API_KEY	Your key from Step 1
LINKEDIN_CLIENT_ID	From your LinkedIn app (Step 2)
LINKEDIN_CLIENT_SECRET	From your LinkedIn app (Step 2)
LINKEDIN_REDIRECT_URI	http://127.0.0.1:8585/callback - keep as-is

Optional - customise your content. Copy each example file and fill it with your own material:

```
cd config cp settings.example.json settings.json cp identity.example.txt identity.txt cp
syllabus.example.txt syllabus.txt cp rules.example.txt rules.txt cp today_notes.example.txt
today_notes.txt
```

- `identity.txt` - who you are (your intro for posts).
- `syllabus.txt` - the topics you know best.
- `rules.txt` - your personal posting rules.
- `today_notes.txt` - today's raw notes; this is what gets turned into a post.
- `settings.json` - schedule, timezone, hashtags, length limits.

■■ **Important:** The real .env and config files are gitignored (never uploaded) - exactly as they should be. Only the .example files are shared. Never commit or share your real ones.

Step 5 - Connect Your LinkedIn Account

STEP 5

Generate an access token (one command)

This connects the tool to your LinkedIn profile, with your explicit approval:

```
npm run auth
```

- The script prints a login URL - open it in your browser.
- Sign in to LinkedIn and approve the **w_member_social** permission.
- Your browser lands on the redirect page with a long address like `http://127.0.0.1:8585/callback?code=...`
- Copy the **entire** address from the address bar and paste it back into the terminal, then press Enter.
- The script prints your token: `LINKEDIN_ACCESS_TOKEN=...`
- Add that line to your `.env` file.

```
# Example value (yours will be unique)
LINKEDIN_ACCESS_TOKEN=your_linkedin_access_token_here
```

■■ **Important:** LinkedIn access tokens expire after about 60 days. When that happens, just run `npm run auth` again and update the value. This token can post to your account - keep it private and never share it.

Step 6 - Test on Your Computer First

STEP 6

Verify everything before going live

Run the automated tests (no internet or keys needed):

```
npm test
```

Generate a post **without publishing anything** - the safest possible first run:

```
npm run dry-run
```

Publish one real post to confirm the full pipeline end-to-end:

```
npm run publish
```

■ **Tip:** Dry-run is your friend: it prints the finished post for your review and posts nothing. Run it first, every time you change your notes or rules.

Step 7 - Upload to GitHub and Add Your Secrets

STEP 7

Push the code, then give the cloud its keys

Part A - upload the code (replace your -username):

```
git remote add origin https://github.com/your-username/linkedin-ai-post-automation.git
git push -u origin main
```

Part B - add your secrets. On GitHub, open your repository and go to **Settings -> Secrets and variables -> Actions -> New repository secret**. Add these two secrets with the same values as your .env file:

Secret name	Value to paste
GEMINI_API_KEY	Your Gemini key from Step 1
LINKEDIN_ACCESS_TOKEN	Your token from Step 5

Part C - only for a public repository. Your real config files are gitignored, so in a public repo they must come from secrets instead. Encode each file and add one secret per file:

```
base64 -w0 config/settings.json # paste output as secret SETTINGS_B64
base64 -w0 config/identity.txt # paste output as secret IDENTITY_B64
base64 -w0 config/syllabus.txt # paste output as secret SYLLABUS_B64
base64 -w0 config/rules.txt # paste output as secret RULES_B64
base64 -w0 config/today_notes.txt # paste output as secret TODAY_NOTES_B64
```

- Private repository and you committed the files directly? Skip Part C entirely.
- Public repository? Add the five secrets above or the cloud can't see your content.

■ **Tip:** GitHub encrypts secrets and masks them in logs. After saving, a secret can only be overwritten - never read again. That is a good thing.

Step 8 - Let It Run Automatically

STEP 8

Run it once by hand, then let the schedule take over

Manual run (recommended first time): open **Actions -> LinkedIn AI Post -> Run workflow -> Run workflow**. A new run starts within seconds.

- Open the run and watch the steps - secret values always appear masked.
- A green checkmark means success: a post was published (or it correctly said “not a posting day”).
- A red cross means a step failed - open its log and compare with the Troubleshooting section.

Schedule: the workflow ships with an example schedule of **Monday and Wednesday at 13:00 UTC**. GitHub's clock is in UTC, so first convert to your own timezone. To change the days or time, edit `.github/workflows/linkedin-post.yml`:

```
# Example - every Tuesday and Thursday at 09:30 UTC:
on:
  schedule:
    - cron: '30 9 * * 2' # Tuesday
    - cron: '30 9 * * 4' # Thursday
```

■ **Tip:** On days that are not scheduled, the workflow still runs and exits cleanly without posting - that is expected behaviour, not an error.

Step 9 - Make It Yours (And Go Public, Optional)

Tune the content, then publish if you like

Everything you post is driven by your own files. Spend a few minutes on them:

```
# Example settings.json - your schedule, topics and quality limits { "timezone": "UTC",
"scheduleDays": ["monday", "wednesday"], "topics": { "monday": "Your Monday Topic",
"wednesday": "Your Wednesday Topic" }, "requiredHashtags": ["#LearningInPublic"],
"postMinWords": 50, "postMaxWords": 250, "maxEmojis": 5, "maxHashtags": 10 }
```

- Set requiredHashtags to tags you want in every post.
- Adjust postMinWords / postMaxWords to control post length.
- Fill today_notes.txt each posting day with what you actually learned.

Going public: when you are happy, flip your repository to public so others can see (and fork) your work:

- Go to **Settings -> General -> Danger Zone -> Change repository visibility -> Public**.
- Double-check the Security Checklist below first.

Troubleshooting

Symptom	Likely cause	Fix
npm run auth says a value is missing	Client ID / Secret / redirect URI not set	Fill all four variables in .env, then run it again
redirect_uri_mismatch error	Redirect URL differs between the app and .env	Make it exactly <code>http://127.0.0.1:8585/callback</code> in both places
Workflow fails with 401 / token error	LinkedIn token expired (~60 days)	Run npm run auth again; update the GitHub secret
Publish step rejects a post	The validator caught a problem (e.g. too short, too many hashtags)	A normal fail-safe: the post is skipped or rewritten - check the log and your rules
"Share on LinkedIn" product unavailable	LinkedIn restricted the product on new apps	Dry-run still works; request the product again in your app settings
Workflow says "not a posting day"	Today is not in your schedule	Expected - edit scheduleDays / the cron to match your preference
Secrets not found in the workflow	Secret added under the wrong scope	Use Settings -> Secrets and variables -> Actions (repository-level)
npm install fails	Node.js version too old	Install Node.js 18 or 20 from nodejs.org and retry

Security Checklist

- Never commit .env or your real config files - they are gitignored.
- Keep your LinkedIn Client Secret and access token private at all times.
- Rotate the Client Secret immediately if it was ever exposed (LinkedIn app settings).
- Tokens expire in about 60 days - renew with `npm run auth` and update the GitHub secret.
- In a public repo, put private config in GitHub Secrets (Part C of Step 7), never in git history.

- Before making a repo public, confirm its history never contained secrets - old credentials live forever in git history.

Quick Command Reference

Command	What it does
npm install	Install the project dependencies (run once)
npm test	Run the automated tests
npm run auth	Generate a new LinkedIn access token
npm run dry-run	Generate a post for review - publishes nothing
npm run publish	Publish one real post now (end-to-end check)
npm start	Run the scheduled logic immediately

Happy posting! Remember the golden workflow: notes → dry-run → review → publish.